ATHUL SUNIL KUMAR

Full Stack Software Engineer

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EDUCATION

Mar Athanasius College of Engineering, Kothamangalam **2023 – 2027**
B.Tech, Computer Science Engineering — CGPA: 8.34/10 | Coursework: Data Structures & Algorithms, DBMS, OOP, Machine Learning

EXPERIENCE

Founder & Full Stack Cloud Developer — Tutora (EdTech SaaS) (tutora.org.in) **Feb 2026 – Present**
Next.js, *React 19*, *TypeScript*, *PostgreSQL*, *Prisma*, *OpenAI API*, *AWS*, *GitHub Actions*

- Designed and built a scalable EdTech platform (enrollment, batch scheduling, attendance, role-based access) as sole technical lead; deployed on AWS ECS Fargate with JWT auth and GitHub Actions CI/CD
- Built a context-aware AI tutor using an SSE streaming pipeline and vector-embedding RAG backend, cutting AI response time-to-first-token by 62% to under 400ms

Freelance Full Stack Developer — Independent **Jun 2024 – Feb 2026**
React, *Vite*, *Node.js*, *Express.js*, *MongoDB*, *REST APIs*

- Partnered directly with parish stakeholders to gather and translate evolving requirements into a full-stack records platform for St. Mary's Cathedral, iterating across multiple release cycles; digitized 113 years of paper-based records into a structured MongoDB model, cutting record lookup time by 95%
- Separately built Victory Home Tutors, a centralized MERN platform unifying student, tutor, and academic management workflows, delivered end-to-end as sole developer

Software Engineering / AI Intern — Novolo.ai **Jun 2025 – Jul 2025**
Python, *Machine Learning*, *AI Agents*

- Built and evaluated a sentiment/semantic analysis model and a CNN-based image classifier, covering preprocessing, training, hyperparameter tuning, and evaluation as part of a cross-functional engineering team

AI/ML Intern — AccelerateX **Oct 2024**

- Completed an ML fundamentals internship — data preprocessing, supervised learning, model training, and validation — and qualified for the Novolo.ai internship via a competitive placement test at completion

PROJECTS

Distributed URL Shortener — url-shortener-1-1y3o.onrender.com/ | Next.js, NestJS, PostgreSQL, Redis, Docker, Nginx

- Designed a horizontally scalable, fault-tolerant system with Base62 hash-sharded PostgreSQL, Redis cache-aside, and sliding-window rate limiting to eliminate single points of failure under concurrency and load
- Achieved 3,450 req/sec throughput at 4.2ms p95 latency with a 99.4% cache hit rate, sustaining 100% uptime in degraded mode during simulated database outages

Churn Guard — Customer Churn Prediction Platform — churn-platform-alb-913716606.ap-south-2.elb.amazonaws.com/ | React, Express, FastAPI, CatBoost, Docker, AWS, Agentic AI

- Identified and resolved target data leakage in a CatBoost model, reaching 0.86 ROC-AUC and 84% accuracy; architected as independent microservices deployed on AWS ECS Fargate with GitHub Actions CI/CD
- Built a 4-tool TypeScript agentic layer with a second-pass verifier for root-cause diagnosis, achieving 100% citation grounding with zero observed hallucinations

DalalSight — Stock Trend Forecasting Platform — dalalsight.vercel.app/ | Python, Flask, AutoARIMA, BiLSTM, XGBoost, SciPy

- Built a Level-2 stacked meta-learning pipeline (AutoARIMA, BiLSTM, Random Forest → XGBoost meta-learner) with SciPy-optimized allocation, raising directional accuracy to 68.7% vs. 52.8% baseline across 3-year NIFTY 50 backtests

TECHNICAL SKILLS

Languages: C, C++, Python, TypeScript, JavaScript, SQL, Java
Frontend & Backend: React, Next.js, Node.js, Express.js, FastAPI, Flask, Spring Boot
Distributed Systems & Cloud: Microservices, Load Balancing, Caching, Sharding, Message Queues (Kafka), AWS (ECS, Fargate, ECR, RDS, ALB, CloudWatch), Docker, Kubernetes, CI/CD
Databases: PostgreSQL, MongoDB, Redis, Prisma
AI & ML: RAG, LLM Applications, NLP, LightGBM, CatBoost, Vector Databases, Ensemble Learning

ACHIEVEMENTS & CERTIFICATIONS

NASA Space Apps Challenge — People's Choice Award (2023) | Amazon ML Challenge (2026) — Reached submission round | 200+ LeetCode problems solved
IBM: Machine Learning with Python, Python for Data Science AI & Development, Introduction to AI, Generative AI: Introduction & Applications
University of Pennsylvania (Wharton, via Coursera): Business and Financial Modeling Specialization